Global data that turn allowed geometry into exact Einstein existence

24 September 2026. This continuation advances three separate existence problems left by the previous development. The first detects global bundle topology from an input metric; the second solves the full deformation problem in a growing-cover family with genuine kernels; the third determines the whole topology of a singular filling from an actual harmonic form. Each has a complete English proof below.

1. Ricci control can force finite monodromy, not merely small local deformation

The cycle theorem concerns an oriented genus-f surface bundle over a closed surface of genus at least two. Horizontal transport of the actual vertical metric defines the accumulated conformal deformation $\mathscr{J}_g(\gamma)$ around a based loop. If a single reference fiber has hyperbolic conformal systole at least ι, then

$$\max_{\gamma \text{ in a generating set}} \mathscr{J}_g(\gamma) < \varepsilon_{f,\iota} \quad \Longrightarrow \quad \begin{array}{c} \text{finite mapping-class monodromy,} \\ \text{exact negative KE metric on the same smooth total space.} \end{array}$$

Properness on the thick part of Teichmüller space forces the generating classes into **one finite group**. Nielsen realization and surface-bundle classification then construct the metric. Finite monodromy is a conclusion, not an assumption. The theorem needs no small Ricci error once the cycle condition is known.

The Ricci composition supplies that condition directly for a Riemannian surface submersion with minimal fibers and hyperbolic base. O'Neill's definite horizontal Ricci identity gives $\|S_X\| \leqslant \sqrt{\delta_-}|X|$. Short generators of the base give the sufficient condition

$$2\,\mathrm{diam}(B)\sqrt{\delta_-(g)} < \varepsilon_{f,\iota}. \tag{1}$$

The fiber metrics need not themselves be hyperbolic. There is no bound on total-space diameter, injectivity radius or volume. The proof also allows an arbitrary base metric, adding $\|(K_B+1)_+\|_\infty$ to δ_- in (1).

This completes an actual pointwise-input-to-Einstein-output implication for the stated submersion class, including the zero-signature case where the older signature obstruction says nothing. It does not assert that an arbitrary near-Einstein metric carries this global fibration.

Two actual Dehn-twist families distinguish the missing information:

- A thick fiber with a long base collar has $\delta_- \to 0$ and two globally closed, normalized opposite forms with $\|\nabla\omega_{\pm}\|_\infty \to 0$, but retains a nonzero accumulated cycle deformation.
- A pinched fiber over a fixed base has the same local smallness and $\mathscr{J}_g \to 0$, but the reference conformal thickness vanishes.

Both lie in the actual bilateral monopole near-saturation regime and have odd first Betti number, hence no Kähler metric. Neither is a proof of general Einstein nonexistence. They show that even exact closedness and uniform near-parallelism do not replace global cycle and moduli control.

2. All Teichmüller modes can be retained and solved, with a sharp scale loss

The full-kernel theorem uses $M_m = B_m \times F$, where B_m is a degree-m cyclic cover of a fixed hyperbolic surface and F is fixed. The exact reference metric is $h_m = b_m + f$. Its gauged Einstein operator has the full kernel $T_{B_m} \oplus T_F$, whose dimension grows with m.

The note proves a uniformly controlled family $H_m(k)$ integrating **every** kernel direction. It proves the remaining equations vanish on that actual family, rather than deleting their projection. For every tensor perturbation in harmonic gauge with $\|g - h_m\|_{C^{3,\alpha}_{\mathrm{uloc}}} < cm^{-2}$,

$$\|g - H_m(\Pi_m(g - h_m))\|_{C^{3,\alpha}_{\mathrm{uloc}}} \leqslant Cm^2 \|\operatorname{Ric}_g + g\|_{C^{1,\alpha}_{\mathrm{uloc}}}. \tag{2}$$

The constants are independent of cover degree. The proof controls the growing-rank TT projection, establishes the scalar Green estimate in the actual supremum norm, and retains the complete tensor decomposition.

The loss in (2) is not a weakness removable by better bookkeeping. Actual smooth metrics satisfy

$$\|\operatorname{Ric}_{g_m} + g_m\|_{C^{1,\alpha}_{\mathrm{uloc}}} \leqslant Cm^{-4}, \qquad \operatorname{dist}_{C^{3,\alpha}_{\mathrm{uloc}}}(g_m, \{\operatorname{Ric}_h = -h\}) \geqslant cm^{-2}. \tag{3}$$

The distance is to **all** normalized Einstein metrics, including diffeomorphic pullbacks, not only a selected product family. A slow harmonic-circle function multiplying a fiber Teichmüller deformation produces (3). Local geometry remains uniformly bounded and noncollapsed.

The prior positive-coercivity repair theorem genuinely excluded these kernels. The new result solves them in this family, but the inverse must pay for slow spatial coefficients even after every exact zero mode is integrated. These manifolds already admit Einstein metrics; the result quantifies repair and its failure of uniformity, not new smooth topology.

3. An actual harmonic form can determine the whole singular filling

The harmonic/contact theorem starts with the previous bounded-diameter gauge-saturated orbifold limit. Let H_j be the self-dual harmonic representative of the actual class projection. The new analytic step upgrades its known mean alignment to

$$-2\pi H_j \longrightarrow \omega_\infty \quad \text{in } C^{1,\beta}$$

on every fixed regular collar. Thus the actual boundary of each whole singular filling carries the canonical **strong convex** contact structure.

If H_j is zero-free on that whole filling, and either the limiting singularity is A_1 with filling $b_2 = 1$, or the filling is a rational homology ball, classical symplectic filling theory determines its smooth type. In the rational-ball case it also forces the singularity to be Wahl. Explicit extension of every possible boundary mapping class then proves the **relative, actually marked Milnor matching** previously assumed. This applies to the whole filling, with any internal bubble scales.

A global Q-Gorenstein smoothing now gives a negative KE metric on the original smooth M_j. The remaining assumptions are actual zero-freeness in the cores, the stated whole-filling homology, and global smoothability. Small averaged deficit has not been asserted to imply them. Conversely a non-Wahl singular point filled by a rational homology ball forces a zero of this specified harmonic form in its core.

4. What is now different about the first line

The advances concern three different pieces of global information:

Information	New completed implication	Remaining unrestricted problem
Cycle transport and thick fiber moduli	Detect finite monodromy and realize an exact metric; in the minimal-submersion class Ricci error supplies the cycle estimate	Produce or replace the global fibration for arbitrary inputs
Full moduli and slow coefficients	Solve all kernel equations and quantify the optimal linear scale loss on cyclic products	Treat general transition data and low-mode systems
Actual harmonic form across a singular core	Force the whole marked filling and return a smoothing to the original smooth type	Control core zeros and global smoothing from weaker inputs

These are additions to the existing integral obstruction theory, not its replacement and not three new claims of worldwide priority. Classical results used in each connection are credited in the proofs. The overall Yau26 assertion remains open here: no theorem above covers every allowed pointwise input. Each existence theorem nevertheless completes the original output requirement in its own stated input class.